```
2. Factorial
3. Quadratic Equation
4. Exit
Enter your choice: 1
Enter a and b: 2 -8
x = 4.00

--- MENU ---
1. Linear Equation
2. Factorial
3. Quadratic Equation
4. Exit
Enter your choice: 1
Enter a and b: 0 5
No solution

--- MENU ---
1. Linear Equation
2. Factorial
3. Quadratic Equation
4. Exit
Enter your choice: 2
Enter an integer (0 to 12): 5
Factorial = 120

--- MENU ---
1. Linear Equation
2. Factorial
3. Quadratic Equation
4. Exit
Enter your choice: 2
Enter an integer (0 to 12): 0
Factorial = 1
```

```
--- MENU ---
1. Linear Equation
2. Factorial
3. Quadratic Equation
4. Exit
Enter your choice: 3
Enter a, b and c: 1 -2 1
Two equal real roots
Root = 1.00

--- MENU ---
1. Linear Equation
2. Factorial
3. Quadratic Equation
4. Exit
Enter your choice: 3
Enter a, b and c: 1 0 1
No real roots

--- MENU ---
1. Linear Equation
2. Factorial
3. Quadratic Equation
4. Exit
Enter your choice: 7
Invalid choice. Try again.
```

```
--- MENU ---
1. Linear Equation
2. Factorial
3. Quadratic Equation
4. Exit
Enter your choice: 3
Enter a, b and c: 1 -3 2
Two distinct real roots
Root 1 = 2.00
Root 2 = 1.00
```